NATIONAL GBV CRISIS TECHNOLOGY • SOUTH AFRICA • FINAL MVP

# (Sankofa Safe)

A dual-persona discreet safety shield engineered for South African GBV survivors • FNB AOTY 2026 Final MVP. Disguised as PowerCalc SA (Municipal Prepaid Electricity & STS Token Calculator) or a store-themed launcher, masking a hardware-encrypted domestic abuse evidence vault and zero-latency emergency beacon.

### PowerCalc SA Camouflage

Runs as an authentic prepaid electricity calculator (Ekurhuleni/Boksburg, City Power, Eskom). 0ms panic exit hides all traces. Zero recents card.

### Life-Safety Beacon

One-touch 72-hour SOS dispatch via SMS/WhatsApp with battery & GPS coordinates telemetry even on R0.00 airtime balance.

### Court-Admissible J88 Vault

AES-256-GCM encrypted incident diary with ECTA SHA-256 hashes, compliant with SA Domestic Violence Act Form 2.

## HUMAN-CENTRED SAFETY UX • ADVERSARY EVASION

End-to-end user story illustrating covert incident logging and critical moment escalation under intimate partner surveillance:

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| **PHASE 1**<br>**Everyday Use** | **PowerCalc SA Facade (Prepaid Electricity Utility):**<br>Survivor uses her phone normally. The app presents as 'PowerCalc SA' with an electric power bolt icon, computing kWh units and STS tokens for Ekurhuleni/Eskom. If inspected by an abusive partner, it appears to be an indispensable household utility. |
| **PHASE 2**<br>**Secret Entry** | **Biometric Evidence Diary & J88 Capture:**<br>In private, survivor enters secret calculation (1234=, 8426=, or 0000=) or long-presses the STS Token button. BiometricPrompt authenticates fingerprint. She logs verbal threats, photos of bruising, and SAPS OB numbers—encrypted on-device with AES-256-GCM. |
| **PHASE 3**<br>**Threat Alert** | **Immediate Threat Escalation (Armed SOS):**<br>A violent altercation begins. Survivor triggers the Quick-Access SOS. A 3-second tactile vibration countdown engages. If not cancelled, high-priority emergency SMS alerts with 72-hour live GPS tracks and battery telemetry dispatch to 5 guardians. |
| **PHASE 4**<br>**0ms Camouflage** | **Zero-Trace Instant Panic Camouflage:**<br>Footsteps approach. Survivor hits the hardware Back button or Escape key. Safe Haven closes in **0 milliseconds**, scrubbing decrypted data from RAM and reverting to PowerCalc SA without task thumbnails. |

## ENGINEERING & ARCHITECTURE SPECIFICATIONS

Modularized Jetpack Compose & React 19 architecture using Manual DI Container and Injectable Services:

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|---|---|
| **UI Layer (Compose & React 19)** | MainActivity • MunicipalPrepaidCalculator • LauncherFacadeScreen • SafeHavenScreen<br>Declarative UI driven by StateFlow and React. PowerCalc SA utility facade, store-themed launcher, and live UAT banner. Zero XML. |
| **Domain & ViewModel (MVVM)** | SafetyTriggerViewModel • SankofaViewModel • UatFeedbackManager<br>Finite state machine managing SOS countdown, cancellation grace period, duress PIN 9999= redirection, and UAT feedback sync. |
| **Dependency Injection (AppContainer)** | AppContainer (Singleton) • SankofaApplication.container<br>Lightweight, manual dependency injection container providing single instances of DAOs, hardware managers, and telemetry helpers with zero runtime reflection. |
| **Data, Security & Cloud Sync** | VaultEncryptionHelper (AES-256-GCM) • Room SQLite / IndexedDB • Vercel & Supabase<br>AndroidKeyStore StrongBox/TEE keys, ECTA SHA-256 hashes, SMS dispatch engine, and Supabase RLS anonymous UAT feedback persistence. |

## LIVE DEMONSTRATION, UAT TESTING & EVALUATION ACCESS

Pre-configured credentials to evaluate PowerCalc SA camouflage unlock, biometric vault, and emergency lifelines:

### PowerCalc SA Secret Triggers

- **Calculator PINs:** 1234=, 8426=, or 0000=
- **Secret Gesture:** Long-press on STS Token button
- **Duress Coercion Alarm:** PIN 9999= (fake voucher + silent SOS)
- **Panic Reset:** Hardware Back key, Escape, or Quick-Exit

### UAT Field Test & Emergency Lifelines

- **UAT Simulated Guardians:** 082 000 0001 / 082 000 0002
- **GBV Command Centre:** 0800 428 428 (Toll-Free, 24/7)
- **SAPS Police Flying Squad:** 10111
- **Zero-Airtime USSD Callback:** *120*7867# / *134*7355#

### Vercel Cloud Deployment & Direct Mobile Downloads

- **Web PWA Hosted Access:** Desktop PC/Mac & iOS Safari zero-trace portal on Vercel Edge
- **Direct Mobile APK:** web/public/downloads/Umkhuseli_Safe_latest.apk (direct clean download)
- **UAT Testing Mode:** Built-in toggle intercepts live SOS to simulate police dispatch without escalating

## LEAN STARTUP METHODOLOGY • PUBLIC BENEFIT MODEL

Social enterprise strategy addressing South Africa's femicide crisis via sustainable, zero-trace technology:

| | | |
|---|---|---|
| **Problem**<br>• SA femicide 5x global avg.<br>• Abusers inspect phones; safety apps cause retaliatory attacks.<br>• Handwritten evidence dropped by court.<br>• Data poverty prevents cloud apps. | **Solution**<br>• PowerCalc SA prepaid utility facade.<br>• Offline AES-256-GCM diary & ECTA hashes.<br>• One-touch 72-hr GPS beacon.<br>• Auto-generates Form 2 Court Affidavits.<br>• Direct clean-domain APK downloads. | **Unique Value Proposition**<br>*'Safety without suspicion; evidence without leakage.'*<br><br>Dual-persona covert protection engineered for South African legal & socio-economic realities. |
| **Unfair Advantage**<br>• Zero-airtime USSD (*120*7867#).<br>• Proprietary 0ms camouflage exit.<br>• Domestic Violence Act Form 2 & J88 medical exam prep integration.<br>• Built-in UAT device testing suite. | **Channels & Distribution**<br>• Thuthuzela Care Centre referrals.<br>• University campus safety offices.<br>• Police forums (CPFs) & NGOs.<br>• Vercel clean web portal distribution. | **Key Metrics & Sustainability**<br>• 0ms camouflage trigger rate (100%).<br>• SOS delivery reliability (>99.5%).<br>• 238/238 automated tests passing.<br>• Public grants & Corporate CSI. |

## SECURITY SOFTWARE DEVELOPMENT LIFECYCLE • POPIA COMPLIANCE

Engineered according to OWASP Mobile Top 10 and South African POPIA & Domestic Violence Act mandates:

### Cryptographic Protections

- **Key Generation:** 256-bit AES keys inside Android Hardware Keystore (StrongBox / TEE).
- **Cipher Mode:** Authenticated AES/GCM/NoPadding with unique 12-byte IVs per record.
- **At-Rest Storage:** Plaintext narratives and injuries never written unencrypted to disk.
- **Tamper-Evidence:** ECTA Section 15 SHA-256 cryptographic hashing.

### Stealth & Memory Hardening

- **Zero-Latency Camouflage:** ViewModel state reset clears decrypted strings in 0 ms.
- **ProGuard / R8 Obfuscation:** Code and symbol obfuscation against decompilation.
- **Zero Third-Party Trackers:** 100% free of advertising SDKs and location telemetry.
- **Section 4(2) Address Withholding:** Complainants can withhold residential address.

### Court Admissibility Guarantee (Act 116 of 1998 & Act 14 of 2021 Form 2 Compliance)

Under the Domestic Violence Amendment Act 14 of 2021 and ECTA Section 15, digital evidence is fully recognized by South African courts. Umkhuseli auto-compiles timestamped incident numbers, perpetrator names, impact summaries, J88 medical notes, and SAPS Occurrence Book references into standardized Form 2 affidavits for immediate submission to the Clerk of the Court.

DEPLOYMENT DELIVERABLES • VERIFIED RELEASE ARTIFACTS

Production-ready Gradle build configuration, verified release artifacts, and deployment pathways:

## Compilation & Build Pipeline

• **Gradle Build Commands:**
./gradlew assembleDebug
./gradlew assembleRelease
./gradlew testDebugUnitTest (79/79 unit & Robolectric tests passing)
• **Compiled APK Artifact:**
web/public/downloads/Umkhuseli_Safe_latest.apk (25.1 MB signed)
• **GitHub Repository:**
github.com/OhhttooeXCIV/Umkhuseli_The_Protector

## Quality Scorecard & Distribution Matrix

• **Target Platform:** Android 7.0 (API 24) to Android 16 (API 36)
• **Web Decoy Platform:** React 19 + TypeScript (159/159 Vitest tests passing)
• **Total Test Coverage:** 238/238 automated tests passing across 25 suites
• **Cloud Deployment:** Vercel Edge configuration with SPA routing
• **Direct Mobile Downloads:** Hosted web APK delivery for clean installs
• **Live UAT Ready:** In-app field testing mode with simulated dispatch intercept